
Learning Accurate and Parsimonious Point Cloud Representations from Images



1 Introduction

My goal is to learn an accurate point cloud representation of a 3D scene directly from 2D images. This is an interesting and challenging problem because point clouds are more flexible and parsimonious compared to other 3D representations like meshes or voxels. In addition, this research has the advantage of being able to be developed in various fields due to the easy-to-edit nature of point clouds. However, rendering and optimizing a point cloud from images is non-trivial due to issues like vanishing gradients when points are far from target pixels.

2 Method

I propose jointly learning a point-based scene representation and a differentiable renderer using only RGB image supervision. The representation will be a point cloud where each point has a position, foreground probability, and feature vector. The renderer will select relevant points for each ray and combine their features to output color. I will design a proximity attention mechanism to address vanishing gradients. By optimizing reconstruction loss between rendered and ground truth images, we can learn point positions and appearance from random initialization.

3 Related Work

Classical point cloud rendering techniques like splatting struggle with vanishing gradients [1]. My attention mechanism will help overcome this. Recent works assume known point clouds from MVS/LiDAR[2]. This model will learn point positions from scratch. Point-NeRF [3] uses slow volumetric rendering. I will use more efficient surface rendering.

4 Plan of Experiments

Implementation will be in PyTorch. The renderer will be a proximity attention module + UNet[6]. This will evaluate on synthetic NeRF dataset [4] and real Forward-Facing Scenes. Also, I will compare image metrics like PSNR, and SSIM [7] to Point-NeRF[5], SPLAT [8], and DPBRF [9].

5 Plan of Project

Since we are a one-person team, I will be doing the entire work. The most difficult part is that the theoretical papers to be used for both point cloud and NeRF are recent papers and the difficulty of the content is beyond the content of the class. I have to study extra for this part. If rendering or attention design doesn't work, I will focus more on applications by assuming known point clouds. Milestones: Attention design, Renderer design, end-to-end training, evaluation, and applications.

References

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